Arithmetic & Number Sense

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Order of operations

Evaluate $12 + 6 \times (9 - 5) \div 3$.

Brackets first: $9 - 5 = 4$. Then $6 \times 4 = 24$, and $24 \div 3 = 8$. Finally $12 + 8 = 20$.

2 [Foundational] • Target: Order of operations

Evaluate $8 + 2 \times 5^2 - 30 \div 6$.

Powers first: $5^2 = 25$. Then $2 \times 25 = 50$ and $30 \div 6 = 5$. So $8 + 50 - 5 = 53$.

3 [Foundational] • Target: Order of operations

Evaluate $\frac{18 - 3 \times 4}{2} + 7$.

Numerator: $18 - 3 \times 4 = 18 - 12 = 6$. Then $6 \div 2 = 3$, and $3 + 7 = 10$.

4 [Foundational] • Target: Order of operations

Evaluate $(7 - 2)^2 - 3 \times (6 - 4)$.

$(7 - 2)^2 = 5^2 = 25$ and $3 \times (6 - 4) = 3 \times 2 = 6$. So $25 - 6 = 19$.

5 [Foundational] • Target: Powers and roots

Evaluate 3^4.

$3^4 = 3 \times 3 \times 3 \times 3 = 81$.

6 [Foundational] • Target: Powers and roots

Evaluate $\sqrt{144}$.

$12 \times 12 = 144$, so $\sqrt{144} = 12$.

7 [Foundational] • Target: Powers and roots

Evaluate $2^3 \times 2^2$ using the laws of exponents.

$2^3 \times 2^2 = 2^{3+2} = 2^5 = 32$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Powers and roots

Evaluate $\sqrt{81} + \sqrt{16}$.

$\sqrt{81} = 9$ and $\sqrt{16} = 4$, so the sum is $9 + 4 = 13$.

9 [Intermediate] • Target: Working with fractions

Compute $\frac{3}{4} + \frac{5}{6}$.

LCD $= 12$: $\frac{9}{12} + \frac{10}{12} = \frac{19}{12} = 1\frac{7}{12}$.

10 [Intermediate] • Target: Working with fractions

Compute $\frac{7}{8} - \frac{1}{3}$.

LCD $= 24$: $\frac{21}{24} - \frac{8}{24} = \frac{13}{24}$.

11 [Intermediate] • Target: Working with fractions

Compute $\frac{2}{3} \times \frac{9}{10}$.

$\frac{2 \times 9}{3 \times 10} = \frac{18}{30} = \frac{3}{5}$.

12 [Intermediate] • Target: Working with fractions

Compute $\frac{5}{6} \div \frac{5}{9}$.

Multiply by the reciprocal: $\frac{5}{6} \times \frac{9}{5} = \frac{45}{30} = \frac{3}{2}$.

13 [Intermediate] • Target: Working with fractions

Which is larger, $\frac{5}{7}$ or $\frac{4}{5}$?

Cross-multiply: $5 \times 5 = 25$ and $4 \times 7 = 28$. Since $28 > 25$, $\frac{4}{5}$ is larger.

14 [Intermediate] • Target: Decimals

Compute $3.45 + 2.7$.

Align decimal points: $3.45 + 2.70 = 6.15$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Decimals • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

Compute $8.6 - 3.25$.

Align decimal points: $8.60 - 3.25 = 5.35$.

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16 [Challenging] • Target: Decimals • Cross-Lesson: 3.01 Algebra Essentials

Compute 1.2×0.5.

$12 \times 5 = 60$, and there are 2 total decimal places, so $1.2 \times 0.5 = 0.60 = 0.6$.

17 [Challenging] • Target: Decimals • Cross-Lesson: 4.01 Geometry & Measurement

Compute $7.5 \div 0.25$.

Multiply both by 100: $750 \div 25 = 30$.

18 [Challenging] • Target: Place value and order of magnitude • Cross-Lesson: 5.01 Quantitative Comparisons

Write $4{,}500{,}000$ in the form $a \times 10^n$ with $1 \leq a < 10$.

$4{,}500{,}000 = 4.5 \times 10^6$.

19 [Challenging] • Target: Place value and order of magnitude • Cross-Lesson: 6.01 Sequences & Patterns

Which is greater, 6×10^4 or $58{,}000$?

$6 \times 10^4 = 60{,}000$, and $60{,}000 > 58{,}000$, so 6×10^4 is greater.

20 [Challenging] • Target: Place value and order of magnitude • Cross-Lesson: 7.01 Data Analysis & Probability

Order from least to greatest: 0.6, $\frac{2}{3}$, 0.65.

$\frac{2}{3} \approx 0.667$. Ordered: $0.6 < 0.65 < \frac{2}{3}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Comparing fractions and decimals

Which value is largest: the point shown on the number line, $\frac{3}{5}$, or 0.55?

a. The marked point (0.7) ✓

b. $\frac{3}{5}$

c. They are all equal

d. 0.55

The marked point is at 0.7. Since $\frac{3}{5} = 0.6$ and $0.55 < 0.6 < 0.7$, the marked point is largest.

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22 [Foundational] • Target: Decimal operations

The number line marks the value of $9.2 - 3.45$. Where does it land?

a. 5.75 ✓

b. 6.25

c. 5.85

d. 5.65

Align decimal points: $9.20 - 3.45 = 5.75$.

23 [Foundational] • Target: Scientific notation & place value

The bar chart shows four city populations. Which one equals 7×10^3?

a. $7{,}000$ ✓

b. $6{,}800$

c. $6{,}500$

d. $7{,}300$

$7 \times 10^3 = 7{,}000$, matching the second bar exactly.

24 [Foundational] • Target: Absolute value & signed numbers

Two of the four points on the number line represent a pair of numbers that are equal in absolute value but opposite in sign. Which pair?

a. -2 and 2

b. -4 **and** 4 ✓

c. -2 and 4

d. -4 and 2

-4 and 4 are the same distance from 0 (4 units) on opposite sides, so they are opposites.

25 [Foundational] • Target: GCD, LCM & factors

The bar chart shows a gardener's bulb counts. They want to plant them in identical rows with the largest possible number of bulbs per row, each row containing only one type of flower. How many bulbs per row?

a. 4

b. 24

c. 6

d. 12 ✓

The largest common row size is the GCD of 36 and 48: $36 = 2^2 \times 3^2$, $48 = 2^4 \times 3$, GCD $= 2^2 \times 3 = 12$.

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26 [Foundational] • Target: Divisibility rules

The bar chart shows four candidate numbers. Which one is divisible by both 5 and 6?

a. 690 ✓

b. 650

c. 685

d. 696

Divisible by 5: ends in 0 or 5. Divisible by 6: even and digit sum divisible by 3. Only 690 satisfies both.

27 [Foundational] • Target: Estimation & rounding

The number line shows the value x rounded to the nearest whole number. If $x = 6.7$, where does the rounded value land?

a. 6

b. 6.5

c. 8

d. 7 ✓

6.7 is closer to 7 than to 6 (since $.7 > .5$), so it rounds to 7.

28 [Foundational] • Target: Number properties

The number line marks 85 between consecutive multiples of 9. Find the remainder when 85 is divided by 9.

a. 3

b. 4 ✓

c. 5

d. 76

$9 \times 9 = 81$, and $85 - 81 = 4$. The remainder is 4.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Order of operations

Evaluate $20 - 4 \times 3 + 6 \div 2$.

a. 11 ✓

b. 27

c. 9

d. 17

Multiply/divide first: $4 \times 3 = 12$, $6 \div 2 = 3$. Then $20 - 12 + 3 = 11$.

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30 [Intermediate] • Target: Order of operations

Evaluate $(5 + 3)^2 - 2 \times 7$.

a. 78

b. 36

c. 50 ✓

d. 46

Brackets and power first: $(5 + 3)^2 = 8^2 = 64$. Then $2 \times 7 = 14$, so $64 - 14 = 50$.

31 [Intermediate] • Target: Order of operations

Evaluate $\frac{45}{3 + 6} \times 2$.

a. 10 ✓

b. 90

c. 7

d. 15

Brackets first: $3 + 6 = 9$. Then $45 \div 9 = 5$, and $5 \times 2 = 10$.

32 [Intermediate] • Target: Order of operations

Evaluate $3 + 2 \times (8 - 5)^2$.

a. 21 ✓

b. 45

c. 17

d. 33

Brackets, then power: $(8 - 5)^2 = 3^2 = 9$. Then $2 \times 9 = 18$, so $3 + 18 = 21$.

33 [Intermediate] • Target: Order of operations

Evaluate $100 - 3 \times (4 + 2 \times 3)$.

a. 64

b. 91

c. 88

d. 70 ✓

Inside the brackets, multiply first: $2 \times 3 = 6$, so $4 + 6 = 10$. Then $3 \times 10 = 30$, and $100 - 30 = 70$.

34 [Intermediate] • Target: Exponents and roots

Use the laws of exponents to evaluate $3^2 \times 3^4$.

a. 243

b. 81

c. 729 ✓

d. 27

$3^2 \times 3^4 = 3^{2+4} = 3^6 = 729$.

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35 [Intermediate] • Target: Exponents and roots

Evaluate $5^3 - 4^2$.

a. 109 ✓

b. 121

c. 93

d. 45

$5^3 = 125$ and $4^2 = 16$, so $125 - 16 = 109$.

36 [Intermediate] • Target: Fraction operations

Compute $\frac{1}{3} + \frac{1}{4} + \frac{1}{6}$.

a. $\frac{1}{2}$

b. $\frac{2}{3}$

c. $\frac{3}{13}$

d. $\frac{3}{4}$ ✓

LCD $= 12$: $\frac{4}{12} + \frac{3}{12} + \frac{2}{12} = \frac{9}{12} = \frac{3}{4}$.

37 [Intermediate] • Target: Scientific notation & place value

Evaluate 3.4×10^4 as an ordinary number.

a. 3,400,000

b. 3,400

c. 340,000

d. 34,000 ✓

Move the decimal point 4 places right: $3.4 \times 10^4 = 34{,}000$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Scientific notation & place value • Cross-Lesson: 5.01 Quantitative Comparisons

In the number 528,437, what is the place value of the digit 8?

a. Hundred thousands

b. **Thousands** ✓

c. Ten thousands

d. Hundreds

Counting from the right: 7 (ones), 3 (tens), 4 (hundreds), 8 (thousands).

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39 [Challenging] • Target: GCD, LCM & factors • Cross-Lesson: 6.01 Sequences & Patterns

Find the GCD of 54 and 72.

a. 9

b. 6

c. 18 ✓

d. 36

$54 = 2 \times 3^3$ and $72 = 2^3 \times 3^2$. Common factors: $2 \times 3^2 = 18$.

40 [Challenging] • Target: GCD, LCM & factors • Cross-Lesson: 7.01 Data Analysis & Probability

Find the LCM of 9 and 15.

a. 3

b. 24

c. 45 ✓

d. 135

$9 = 3^2$, $15 = 3 \times 5$. LCM $= 3^2 \times 5 = 45$.

41 [Challenging] • Target: GCD, LCM & factors • Cross-Lesson: 8.01 Word Problems & Reasoning

How many factors does 60 have?

a. 6

b. 8

c. 12 ✓

d. 10

$60 = 2^2 \times 3 \times 5$. Number of factors $= (2+1)(1+1)(1+1) = 12$.

42 [Challenging] • Target: GCD, LCM & factors • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

Is 119 a prime number?

a. **No —** $119 = 7 \times 17$ ✓

b. Yes, it is prime

c. No — $119 = 11 \times 9$

d. Cannot be determined

$119 \div 7 = 17$, so $119 = 7 \times 17$ has factors other than 1 and itself.

43 [Challenging] • Target: Divisibility rules • Cross-Lesson: 3.01 Algebra Essentials

Without dividing, is $5{,}481$ divisible by 3?

a. Cannot tell without dividing

b. **Yes** ✓

c. No

d. Only by 9, not 3

Digit sum: $5 + 4 + 8 + 1 = 18$, divisible by 3 (and by 9), so $5{,}481$ is divisible by 3.

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44 [Challenging] • Target: Estimation & rounding • Cross-Lesson: 4.01 Geometry & Measurement

Round 147.362 to the nearest tenth.

a. 147.3

b. 150

c. 147.36

d. 147.4 ✓

The hundredths digit is 6, so round up: 147.4.

45 [Challenging] • Target: Estimation & rounding • Cross-Lesson: 5.01 Quantitative Comparisons

Estimate 312×48 by rounding each factor to the nearest ten, then compare to the exact product $14{,}976$.

a. Estimate $310 \times 50 = 15{,}500$, far from the exact value

b. Estimate $300 \times 50 = 15{,}000$, close to the exact value

c. Estimate $320 \times 40 = 12{,}800$, close to the exact value

d. **Estimate $310 \times 50 = 15{,}500$, close to the exact value** ✓

312 rounds to 310 and 48 rounds to 50: $310 \times 50 = 15{,}500$, reasonably close to $14{,}976$.

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